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Large Language Models in Radiology Reporting—A Systematic Review of Performance, Limitations, and Clinical Implications

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Abstract

Background

Large language models (LLMs) have emerged as potential tools for automated radiology reporting. However, concerns regarding their fidelity, reliability, and clinical applicability remain. This systematic review examines the current literature on LLM-generated radiology reports.

Methods

We conducted a systematic search of MEDLINE, Google Scholar, Scopus, and Web of Science to identify studies published between January 2015 and February 2025. Studies evaluating LLM-generated radiology reports were included. The study follows PRISMA guidelines. Risk of bias was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool.

Results

Nine studies met the inclusion criteria. Of these, six evaluated full radiology reports, while three focused on impression generation. Six studies assessed base LLMs, and three evaluated fine-tuned models. Fine-tuned models demonstrated better alignment with expert evaluations and achieved higher performance on natural language processing metrics compared to base models. All LLMs showed hallucinations, misdiagnoses, and inconsistencies.

Conclusion

LLMs show promise in radiology reporting. However, limitations in diagnostic accuracy and hallucinations necessitate human oversight. Future research should focus on improving evaluation frameworks, incorporating diverse datasets, and prospectively validating AI-generated reports in clinical workflows.

Keywords

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Radiology reports, Large Language Models, Artificial Intelligence, Natural Language Processing, Automated Reporting, Clinical Evaluation, AI Alignment, Generative AI

Introduction

Radiology reports are the main communication interface between radiologists and referring physicians [1]. They can significantly influence medical decisions. However, there are numerous challenges associated with current radiology reporting that require attention.

One challenge in radiology reporting is inter-observer variability, referring to differences in how radiologists describe and interpret the same imaging findings [2, 3]. Different terminology, level of detail, and subjective descriptions may lead to inconsistent reporting. This lack of uniformity can result in miscommunications, delays in patient management, and even diagnostic errors [4]. Additionally, the increasing workload burden on radiologists contributes to fatigue, cognitive errors, and longer report turnaround times [5, 6].

Large language models (LLMs) have emerged as a promising tool for automated radiology reporting. They can promote consistent, structured, and comprehensible reports [7, 8]. However, LLMs also introduce challenges, including biased outputs, hallucinations, and ethical and legal concerns [9, 10]. Ensuring fidelity and trustworthiness of LLM-generated reports is imperative if these models are to be safely integrated into clinical practice.

The aim of this study was to systematically review the current literature on the use of LLMs in generating radiology reports. We discuss their benefits, limitations, and potential future implications.

Methods

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Literature Search

We systematically searched the literature to identify studies describing the generation of radiology reports using LLMs. We searched MEDLINE, Google Scholar, Scopus, and Web of Science for papers published from January 2015 up to February 2025. The full search process including Boolean operators is detailed in the **Supplementary Materials**. In addition, we checked the reference lists of selected publications and the "Similar Articles" feature in PubMed, to identify additional publications. Ethical approval was not required, as this is a systematic review of previously published research and does not include individual participant information. Our study followed the Preferred Reporting Items for Systematic Reviews and meta-analyses (PRISMA) guidelines [11]. The study is registered with PROSPERO (CRD 42025647882)

Study Selection

All search results were imported into a single CSV table and deduplicated. Two authors (YA and VS) independently screened titles and abstracts for relevance. Potentially eligible articles were retrieved in full text and assessed by YA and VS. Discrepancies were resolved by a third author (EK).

Inclusion and Exclusion Criteria

Included were full-text peer-reviewed publications in English focusing on radiology report generation using LLMs. We excluded non-English articles, non-original research, non-peer reviewed, studies that did not assess LLMs, and studies that did not explicitly assessed radiology reports or impressions generation. **Figure 1** presents the flow diagram of the screening and inclusion process.

Quality Assessment

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The risk of bias and applicability was evaluated using the tailored Quality Assessment of Diagnostic Accuracy Studies tool 2 (QUADAS-2). (**Figure 2.**)

Performance Evaluation

LLM performance involved either quantitative or qualitative approaches. Quantitative evaluations assessed similarity between AI-generated and reference reports. These were based on automatic natural language processing (NLP) metrics [12, 13, 14]. Explanations and key differences of these metrics are detailed in **Supplementary Table 1.** Qualitative metrics were based on human evaluation.

Results

Study Selection and Characteristics

Nine studies were included in this review, published between September 2023 and August 2024. Six studies utilized different versions of generative pre-trained transformers (GPT) (66%). Five studies focused on chest X-ray (CXR) reports (55%). One study focused on breast ultrasound reports (11%), one used MRI reports (11%), one used dental X-ray images (11%), and two used CT reports (22%) (**Table 1**). The results of the studies are summarized in **Table 2**. **Figure 3** provides an overview of the characteristics of the included studies.

Non-peer-reviewed publication were excluded from our analysis, but their results are detailed in the **Supplementary Material,** along with A summary presented in **Supplementary Table 2.**

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LLM Performance in Generating Full Radiology Reports (Image-to-Text) vs. Radiology Impressions

Six studies evaluated full report generation based on image analysis. Three studies focused on radiology impression generation based on the findings sections of the reports (**Figure 4**).

Li et al. [15] showed that an LLM enhanced with a knowledge graph outperformed base models in generating radiology reports. They extracted regional image features using a transformer model. These were then combined with disease-specific features from the knowledge graph. Their model outperformed other pre-trained LLMs for radiology reporting. However, its reliance on a constrained knowledge graph limited generalization beyond the scope of chest X-rays and introduced potential biases.

Voinea et al. [16] assessed the fine-tuned Llama 3-8B for full reports generation. Their model produced clinically relevant impressions achieving a ROUGE-1 F1 score of 0.4998, ROUGE-L F1 score of 0.4628, and a BERTScore F1 of 0.4282. While AI-generated conclusions were preferred in 21.8% of cases, radiologists still found that in many other cases, the model's conclusions lacked clinical judgment and exhibited biases due to dataset limitations. Similarly, Sun et al. [17] evaluated GPT-3.5's ability to generate BI-RADS classifications and treatment recommendations from breast ultrasound reports, noting structured output but frequent misdiagnoses, overdiagnoses, and classification inconsistencies (**Table 2, Table 3**).

Lee et al. [18] compared M4CXR and GPT-4 for interpreting chest X-rays and generating associated radiology reports. M4CXR outperformed GPT-4 in diagnostic accuracy and anatomical localization, while GPT-4 produced more false findings and hallucinations. Nakaura et al. [19] showed that GPT-3.5 and GPT-4 achieved high readability in their generated reports, but underperformed in differential diagnoses compared to radiologists.

Stephan et al. [20] found that GPT-generated reports from dental panoramic radiographs (OPGs) had high textual similarity and better readability than student-written reports, but lacked critical diagnostic details.

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Fine-Tuned vs. Out-of-the-Box Model Performance

There were six studies evaluating base LLMs, and three studies evaluated pretrained or fine-tuned models. Li et al. [15] showed that a fine-tuned LLM outperformed base models on NLP metrics (BLEU-4 improved by 4.5%, higher RadGraph-F1, BERTScore, and RadCliQ) (**Table 2**). Additionally, the fine-tuned model demonstrated improved alignment with human reports as measured by NLP metrics, though direct human evaluation was not reported. Soleimani et al. [21] reported that GPT models generated structurally coherent reports but missed subtle findings that radiologists identified. Voinea et al. [16] showed that a fine-tuned Llama 3-8B significantly improved BLEU, ROUGE, METEOR, and BERTScore over base models for full radiology report generation.

Fine-tuning generally improved performance. For example, Zhu et al. [22] showed that a fine-tuned GPT-4 achieved higher correlation with expert evaluations ($\tau = 0.48$) compared to base models**.** Regressed GPT-4, referring to a post-processing approach where a regression model is trained using expert-annotated radiology reports to adjust GPT-4-generated scores, improved alignment further ($\tau = 0.64$). However, fine-tuned models still exhibited hallucinations and inconsistent clinical accuracy when lacking diverse training data (**Table 2, Table 3**).

Automated Metrics vs. Human Expert Evaluation

Banerjee et al. [23] examined text-based evaluation metrics, finding that most automated metrics overemphasized stylistic similarity rather than clinical correctness. The study highlighted FineRadScore-GPT-4 as the only metric that correlated well with expert assessments, primarily due to its focus on clinical relevance and semantic accuracy rather than mere context similarity (**Table 2**).

Banerjee et al. [23] also noted that Natural Language Generation (NLG) metrics, which evaluate textual similarity and fluency using measures such as BLEU, ROUGE, and METEOR,

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correlated poorly with expert assessments, often ranking stylistically similar AI reports higher, regardless of diagnostic accuracy. Specifically, they assessed BLEU-2.

Zhu et al. [20] proposed a detailed few shots (5) GPT-4 evaluation model, which outperformed traditional metrics in aligning with expert radiologists. Metrics like FineRadScore-GPT-4 were better aligned with radiologists' evaluations, while metrics like RadGraph F1 and RadCliQ [24] helped capture clinical relevance (**Table 2**).

Key Limitations of Current AI in Radiology Reports

Across the nine studies reviewed, AI-generated reports exhibited hallucinations (reported in 44% of studies, 4/9), misdiagnoses (55%, 5/9), and missing clinical details (66%, 6/9) as frequent errors. Lee et al. [18] reported that GPT-4 misidentified anatomical locations and over-reported findings. Stephan et al. [20] highlighted AI's struggles with integrating broader clinical context. Sun et al. [17] found GPT-3.5 frequently overdiagnosed benign lesions, and Nakaura et al. [20] showed that GPT models underperformed in differential diagnosis accuracy (**Table 3**). Examples of the various prompts used in the studies are detailed in **Table 4**. Limitations examples mentioned in the studies are presented in **Figure 5 and Figure 6.** Further examples are included in the **Supplementary Material.**

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Discussion

In this systematic review, we explored the role of LLMs in radiology report generation. Several studies showed that these models can improve consistency, readability, and structure. However, they also reveal critical limitations in diagnostic accuracy, clinical reasoning, and attention to nuanced clinical context, where human radiologists demonstrate clear superiority.

Producing a meaningful radiology report requires technical accuracy, clear communication, and clinical synthesis [25]. It must be understandable and actionable for various stakeholders. These include patients, clinicians, and support staff. Notably, none of the included studies examined how clinicians or patients perceived the AI-generated reports, although LLMs could potentially offer clearer, more accessible communication [26, 27].

A recurring theme across the reviewed studies was the variability in LLMs' performance across different tasks. Stylistic bias and overfitting to training data patterns compromised both performance and the validity of the evaluation metrics. Several studies demonstrated that automated metrics, such as BLEU, ROUGE, and METEOR, were disproportionally influenced by stylistic similarities, rather than clinical accuracy [15, 16, 23]. Scores that rely on semantic similarities, such as BERTScore and RadGraph may offer more relevant insights, but these also fall short of alignment with actual clinical relevancy [28]. Human evaluations, on the other hand, are subjective and not scalable for large datasets. This creates a gap in reliable metrics for evaluation.

Several studies also noted the importance of the input quality [16, 17, 20]. This dependence was particularly evident in studies involving multimodal data, such as breast ultrasound or chest X-ray interpretation, where imaging findings alone were insufficient for comprehensive reporting [16, 17, 21]. Collaboration between AI and radiologists may be a promising direction. In studies where radiologists actively refined or corrected AI outputs, report quality improved. Yet, such approaches

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are only practical if they reduce workload rather than increase the burden of validation and correction in the clinical routine [29].

Future research should focus on improving evaluation frameworks, including diverse datasets to mitigate biases, and prospectively validating AI-generated reports in clinical workflows. Real-world validation remains sparse but essential to translate these advances into reliable clinical tools. Numerous commercial radiology reporting software tools are already widely used in radiology practice. Yet their performance, failure modes, and reliability are largely unstudied in the academic literature. Understanding how these systems compare to newer, research-driven models will be essential. Furthermore, with the pace of generative AI development, it remains to be determined how base models compare to LLMs that are fine-tuned on specialized radiology datasets.

Fine-tuning can align a model's output with a specific reporting style. This can improve user satisfaction and clinical relevance [30]. However, this customization may also potentially increase the risk of "automation bias", where radiologists trust these familiar outputs without adequate review. Fine-tuning also exposes training data to adversarial attacks [31]. It also demands resources, including large, high-quality datasets and specialized expertise. Meanwhile, proprietary models continue to improve rapidly, potentially offering more advanced clinical reasoning without explicit domain-specific training [32]. Another factor is test-time computation. That is the computational resources used when the model generates a report. Iterative refinement can potentially yield higher-quality outputs but at greater cost and longer processing times [33, 34]. As such, both fine-tuning and test-time computations will have to be weighed against clinical needs, and resource availability.

Our review has several limitations. Due to heterogeneity in study design and data, a meta-analysis was not performed. The majority of the studies focused on one modality (X-ray) and included limited LLM performance in other imaging modalities, such as CT or MRI. One study did not include a human expert evaluation of the LLMs' performance. We excluded non-English studies, which can limit diverse insight into LLM performance in a global context. Also, most of the studies

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did not include examples of the AI limitations, which could provide insight into real-world integration issues. None of the studies included have evaluated these tools in a real-world hospital workflow. Additional studies in a real-world environment will be needed to explore AI fidelity in radiology reporting.

In conclusion, LLMs may be a valuable tool for radiology reporting, but their responsible deployment requires careful attention to clinical accuracy, transparency, and human oversight. Ensuring alignment between AI and human goals is essential. By integrating AI as an adjunct, future systems can benefit from the efficiency and consistency of LLMs while preserving the clinical reasoning and contextual understanding unique to human expertise.

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Declarations

Ethics approval and consent to participate:

Not applicable

Consent for publication:

Not applicable

Availability of data and materials:

All data generated or analyzed during this study are included in this published article

Competing interests:

The authors declare that they have no competing interests

Funding:

Not applicable

Authors' contributions:

YA conducted the literature search, data synthesis and wrote the primary manuscript. BSG, JC, and PK reviewed and edited the manuscript. GNN, EK, and VS contributed to the conceptualization, data synthesis, and editing of the manuscript. All authors read and approved the final manuscript.

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Figure 1. Flow diagram of inclusion and exclusion process

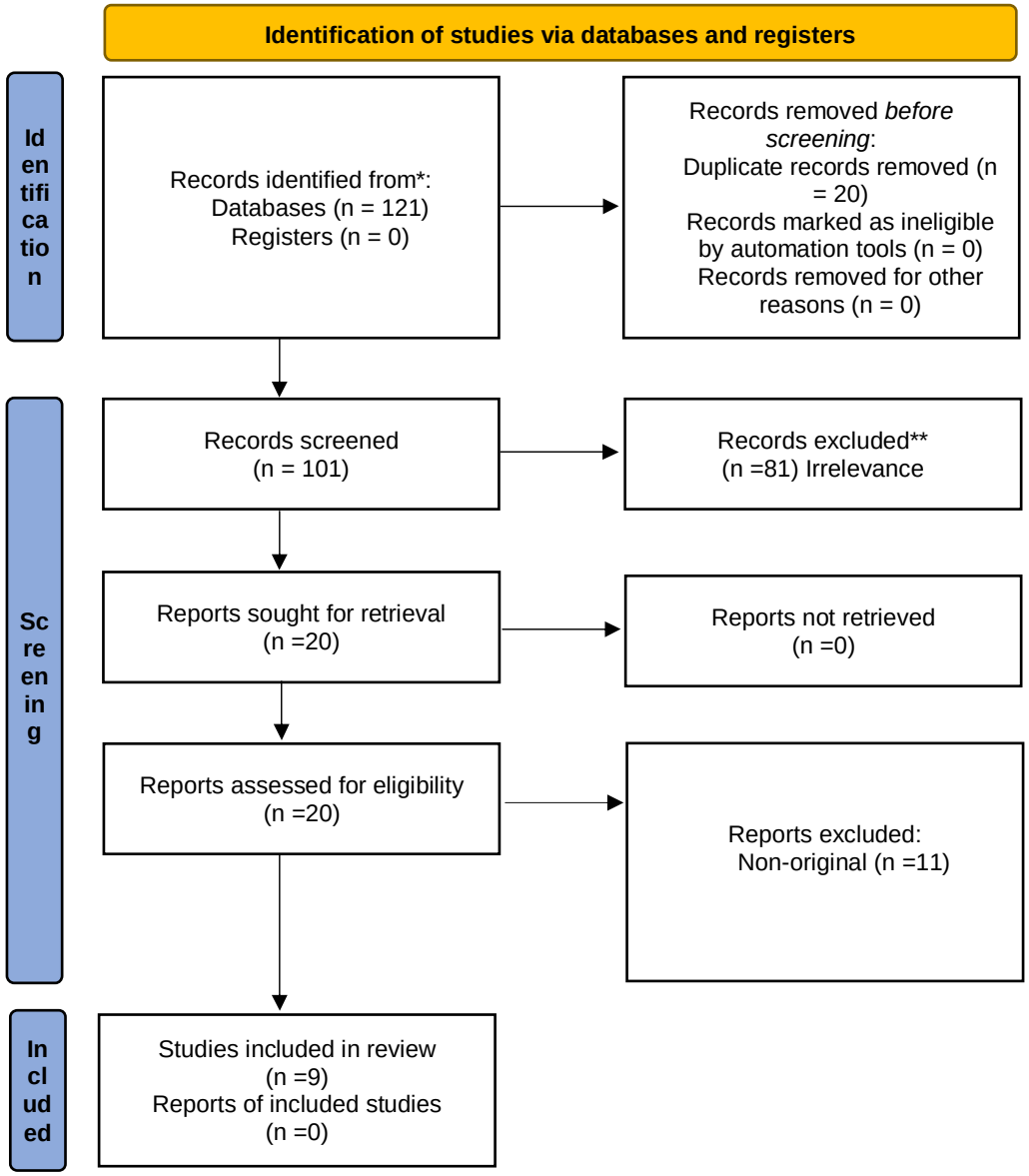

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Figure 2. Risk of bias and applicability assessment using QUADAS-2 tool

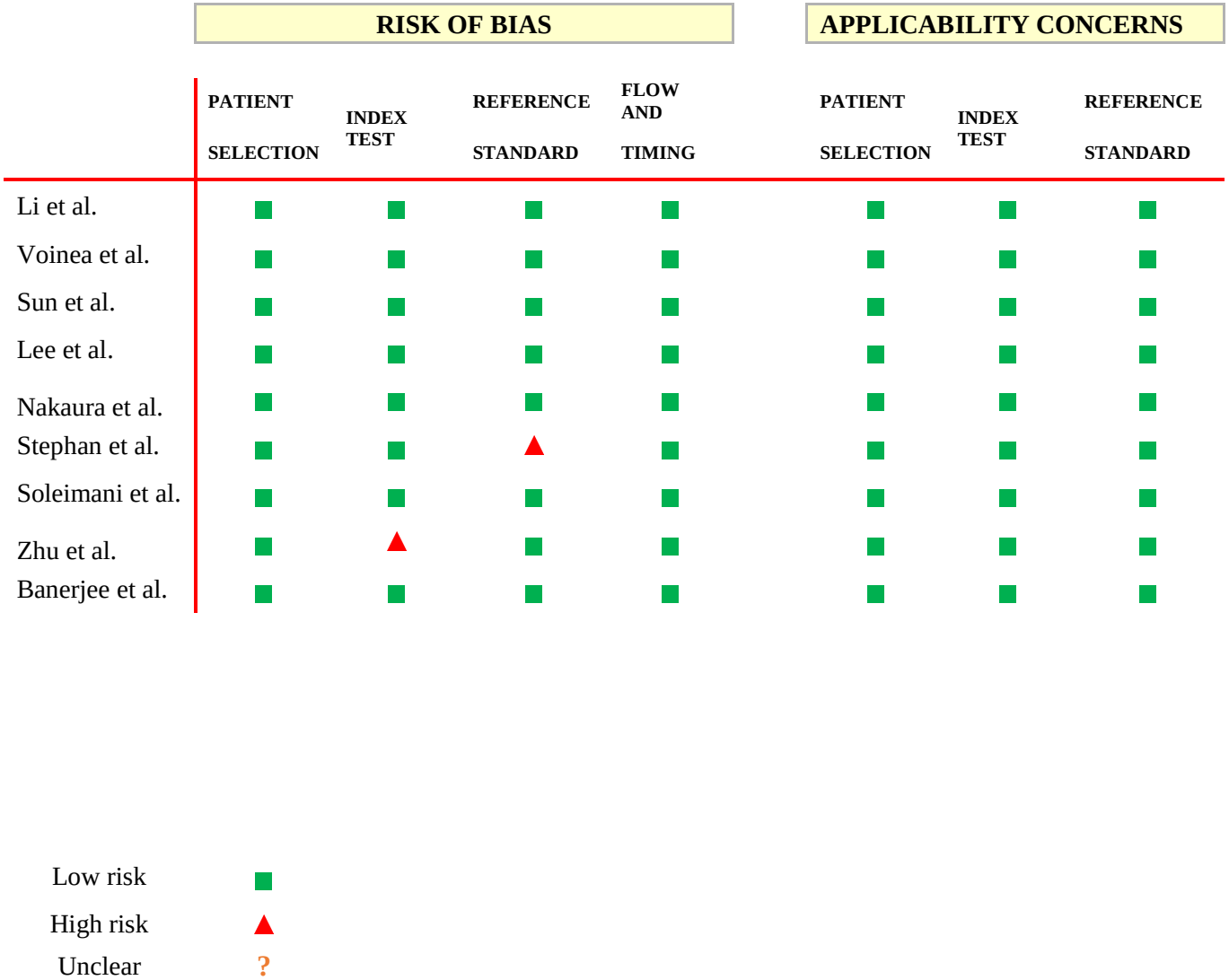

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Figure 3. Overview of differences and similarities across studies

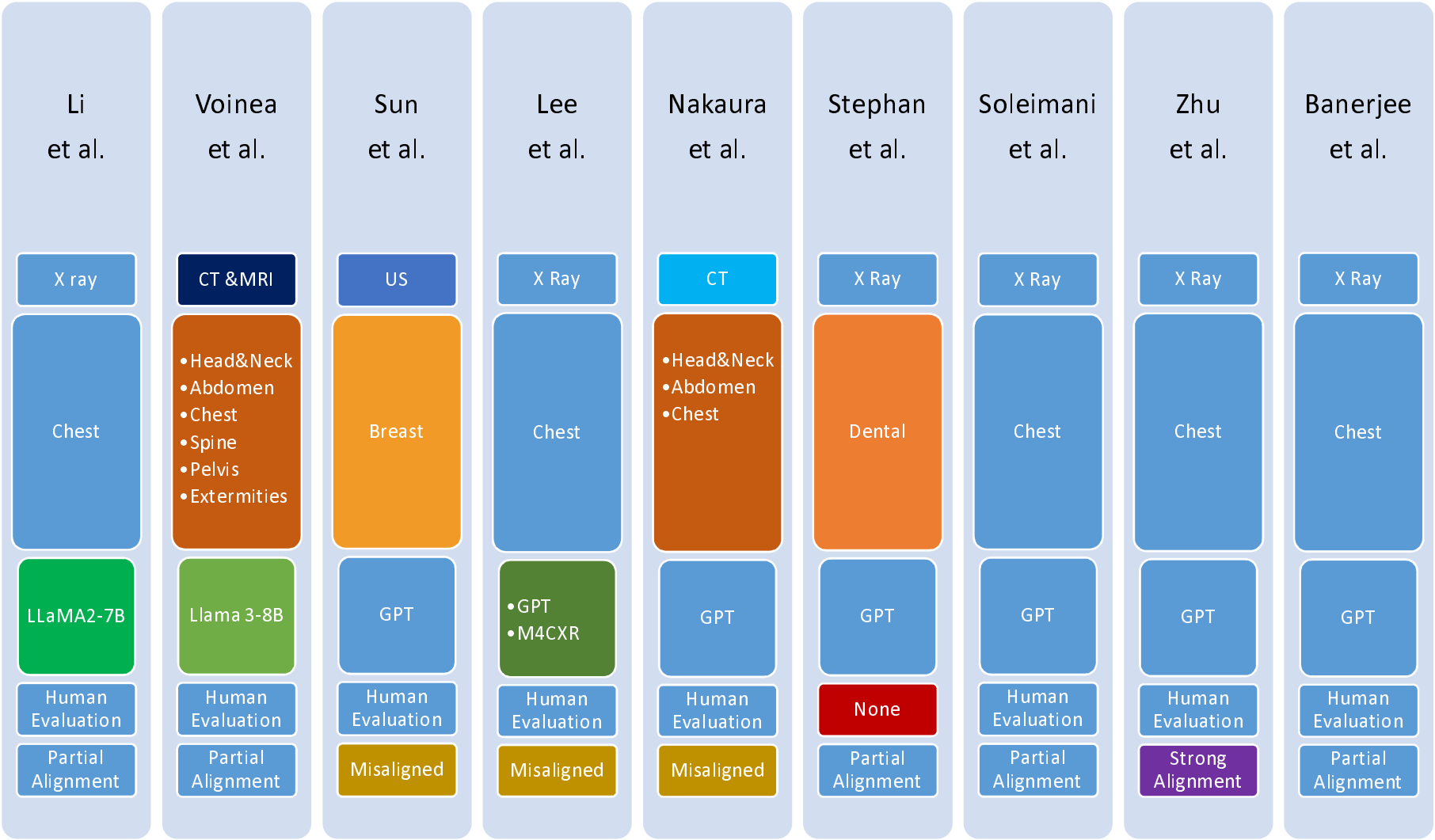

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Figure 4. Schematic illustration demonstrating AI report generation

"In the GPT-2 generated report, a list of differential diagnoses is not even created, and the impression primarily suspects hepatocellular carcinoma. In the GPT-3.5 generated report, although the format is well organized, hepatocellular carcinoma is still listed as the top differential diagnosis:

GPT-2

Findings: There is a 22 mm diameter lesion in segment 8 of the liver. The lesion is Enhancement in delayed phase

Impression: The most likely diagnosis is a hepatocellular carcinoma. Other possibilities include metastatic disease or a benign lesion such as a hemangioma.

Differential Diagnoses:

GPT-3.5

Findings: A contrast enhanced abdominal CT revealed a delayed phase-enhancing lesion measuring 22mm in diameter, located in segment 8 of the liver.

Impression: The delayed phase-enhancing lesion in the liver, along with the patient's age and sex, suggests the possibility of a malignant tumor such as hepatocellular carcinoma or intrahepatic cholangiocarcinoma. It is recommended to perform additional diagnostic tests such as MRI, biopsy or serum tumor marker tests for further characterization and management.

Different Diagnoses:

1. Hepatocellular carcinoma
2. Intrahepatic cholangiocarcinoma
3. Metastatic liver disease
4. Focal nodular hyperplasia

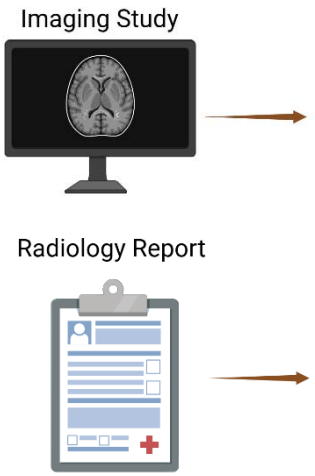

Figure 5. Limitation Example, lower accuracy in differential diagnosis, Nakaura et al.

Figure 6. Limitation Example, lower accuracy in BI-RADS classification and malignancy diagnosis, Sun et al.

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Table 1. General features of reviewed studies

Study	Dataset	LLM	Modality	Alignment
Li et al. [15]	IU-Xray MIMIC-CXR	LLaMA2-7B	Chest X-Ray	Partially aligned (outperformed existing models)
Voinea et al. [16]	21,152 Radiology Reports from the University of Medicine and Pharmacy of Craiova's Imaging Center	Llama 3-8B	MRI CT	Partially aligned (acceptable structure, minor inconsistencies)
Sun et al. [17]	131 Ultrasound Reports from Beijing Friendship Hospital	GPT-3.5	Breast Ultrasound	Misaligned (misclassification errors)
Lee et al. [18]	826 PA Chest Radiographs from Inha University Hospital, Incheon	M4CXR GPT-4o	Chest X-Ray	Misaligned due to hallucinations
Nakaura et al. [19]	28 Images from the Radiology Review Manual, 8th ed	GPT-2 GPT-3.5 GPT-4	CT	Misaligned (missed diagnostic details)
Stephan et al. [20]	Unknown Orthopantomogram (OPG)	GPT (Version not mentioned)	Dental X-Ray	Well-aligned in readability, but lacked diagnostic depth
Soleimani et al. [21]	1000 Records from MIMIC-CXR	GPT (Version not mentioned)	Chest X-Ray	Partially aligned in impression generation
Zhu et al. [22]	MIMI C-CXR	GPT-4 GPT-3.5	Chest X-Ray	Well-aligned with expert assessments
Banerjee et al. [23]	Private Dataset Containing Chest X-ray Reports Worldwide	GPT-4	Chest X-Ray	FineRadScore-GPT-4 was well-aligned, but other metrics failed

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Table 2. Evaluation metrics and key results

Study	Task	Human Evaluation	BLEU	BERTScore	ROUGE	METEOR	Clinical Metrics	Other
Li et al. [15]	Full Report Generation	Expert radiologists (No. Not mentioned)	BLEU-4: **0.140**	High (Exact score not mentioned)	ROUGE-L improved	Best Scores in MIMIC-CXR **(0.165)** and IU-Xray **(0.218)**	RadGraph-F1, RadCliQ, CheXpert Similarity (Best scores)	CIDEr used
Voinea et al. [16]	Full Report Generation	13 Doctors	score of **0.225**	Precision: **0.8054** Recall: **0.7710** F1: **0.786**	ROUGE-1: **0.4998**) ROUGE-L: **0.4628**	**0.4282** (+0.0561)	N/A	**Turing-like Evaluation**: 21.8% of cases preferred AI **Overall rating**: 3.65 out of 5
Sun et al. [17]	Impression Generation	57 Doctors	N/A	N/A	N/A	N/A	BI-RADS Classification (κ = 0.68) Malignancy Diagnosis (p = 0.033)	**Turing Test**: 35.34% of AI reports mistaken for human-written
Lee et al. [18]	Chest X-Ray Interpretati on	Two board-certified radiologists	N/A	N/A	N/A	N/A	M4CXR Accuracy: 60.5% GPT-4 Accuracy: 42.5%	Higher hallucination rates in GPT-4
Nakaura et al. [19]	Differential Diagnosis	One board-certified radiologist One radiology resident	N/A	N/A	N/A	N/A	Top-1 Accuracy: GPT-4 **54%**, GPT-3.5 **54%**, GPT-2 **21%** Radiologists: **100%**	Kappa analysis: Overall quality (κ = 0.42)
Stephan et al. [20]	Dental Radiology Reporting	Dental Experts Radiology instructors	N/A	Precision: **0.967** Recall: **0.958** F1: **0.962**	N/A	N/A	N/A	**Readability** (FRE: p = .49 LIX: p = .09) **Findings Per Spreadsheet** AI included fewer findings (P = .003 and P = .014).
Soleimani et al. [21]	Findings & Impression Generation	N/A	N/A	**Findings Similarity**: Bart **99.3%**,	N/A	N/A	N/A	Cosine Similarity, TF-IDF, Jaccard, Sequence

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				XLM **98.9%** **Impression Similarity**: Word-Embedding **84.4%**, MPNet **75.7%**				Matching, XLM, MPNet
Zhu et al. [22]	Full Report Evaluation	Six radiologists	Lower correlations	N/A	Lower correlations	METEOR ($\tau = 0.29$)	CheXbert, RadCliQ, RadGraph-F1, FineRadScore-GPT-4	Few shot GPT-4 (5-shot) performed best ($\tau = 0.48$)
Banerjee et al. [23]	Metric Evaluation	Internal medicine attending physician Radiology resident	BLEU-2 (worst performance, negative correlation)	BERTScore highly sensitive to stylistic differences	N/A	N/A	FineRadScore-GPT-4 (best metric, no stylistic bias)	RadGraph-F1, CheXpert, RadCliQ

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Table 3. Limitations and drawbacks of AI performance

Study	Limitations				
Li et al. [15]	Limited Scope of the Knowledge Graph	Focus on Chest X-rays Only	Multimodal Fusion Complexity	Dependency on Knowledge Graph Quality	Potential Bias from Training Data
Voinea et al. [16]	Dataset Limitations	Dependence on Input Quality	Lack of Full Clinical Context	Challenges in Complex Cases	Voinea et al. [14]
Sun et al. [17]	Lower Accuracy in BI-RADS Classification and Malignancy Diagnosis	Inconsistencies Between Report Components	Overdiagnosis of Benign Lesions	Text-Only Input Limitation	Reduced Performance on Complex Cases
Lee et al. [18]	Lower Diagnostic Accuracy for GPT4o	Higher Rate of False Findings for GPT4o	Inferior Performance in Location Accuracy for GPT4o	Higher Hallucination Rate for GPT4o	Slower Processing Time for GPT4o
Nakaura et al. [19]	Lower accuracy in differential diagnosis	Incomplete alignment with radiologists' impressions	Hallucination and fabrication of findings	Moderate interobserver agreement	
Stephan et al. [20]	Omission of Critical Diagnostic Information	Challenges in Interpreting Complex Odontogram Data	Dependence on Prompt Design	Potential for Bias Due to Training Data	Variability in Readability
Soleimani et al. [21]	Limited Understanding of Context and Subtle Findings	Inconsistent Performance Across Report Sections	Lack of Ground Truth Interpretation Process	Potential Overfitting to Report Style	Lack of Image Analysis Capability
Zhu et al. [22]	Lower Performance of GPT-3.5 Compared to GPT-4	Stylistic Sensitivity and Bias in Traditional Metrics	Limited Clinical Insight in NLG and CE Metrics	Challenges in Capturing Sentence-Level Nuances	Dependence on Annotated Data from Experts
Banerjee et al. [23]	Sensitivity to Stylistic Differences	Poor Alignment with Expert Judgment	Inconsistent Performance Across Sites		

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Table 4. Various prompt examples

Study	Prompt Examples
Li et al. [15]	"Generate a comprehensive and detailed diagnosis report for this radiology image."
Voinea et al. [16]	N/A
Sun et al. [17]	"Please summarize the BI-RADS classification, benign/malignant, and brief treatment recommendations based on the following ultrasound diagnostic results. The following is a description of breast ultrasound: "Examination findings: Both breasts have uneven echogenicity and disordered structure, with no mass or abnormal blood flow signals observed in the left breast. In the right breast, multiple low-to-no echogenic nodules are found, with the largest one located at the 3 o'clock direction about 0.9cm from the nipple in the glandular layer, measuring approximately 0.7x0.3cm, having clear and regular boundaries, and no evident blood flow signals within. No abnormally enlarged lymph nodes are detected in both axillae."
Lee et al. [18]	"This is a chest PA image. Tell me more about what is going on?"
Nakaura et al. [19]	"...Provide a concise and prioritized catalogue of differential diagnoses without a corresponding disease description, comprising no more than five suspects, ranked by level of suspicion."
Stephan et al. [20]	"Please write a dental radiology report for the following findings derived from an orthopantomogram using appropriate radiological terminology: [Findings list provided here]. Structure the report into a findings section and an impression section. Please follow the style and conventions of radiological reports used in dental education."
Soleimani et al. [21]	"Utilizing the prognoses outlined in Step 1 and the findings acquired in Step 2, compose a radiology report comprising two sections that accurately describe the health condition of the patient who visited the hospital. Please respond only with the information I request. Please provide the information as follows: Section One: Offer an impression in a single paragraph. Section Two: Suggest a diagnosis concisely, without any additional recommendations."
Zhu et al. [22]	"Your task is to evaluate the prediction sentence by sentence, comparing it with the original report. You should also have one overall score from 0-5 for prediction sentences with respect to the original report based on your subjective impression. And I only need the csv format for the result."
Banerjee et al. [23]	"Pretend you are a radiologist and format the content of these notes in a polished findings and impressions section. Your findings section may be long or short. Your impression should only have 1-3 lines. If you are unsure about an abbreviation, term, or other odd phrasing, make your best guess. Match

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	the style of this radiology report…”